Solver Documentation

electroChem

PotentialReaction branch

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1 Introduction

This document covers the custom OpenFOAM solver electroChem made for simulating reacting multicomponent potential flows, with the goal of simulating microbial electrosynthesis of CO_2 to acetate. The solver is an extension of the existing pimpleFOAM FVM solver from OpenFOAM.

1.1 PotentialReaction branch

This branch restructures the code and couples equilibrium reactions and potential corrections to the transport equation ensuring the solution is converged on the transport path. The reaction setup is restructured to treat general equilibrium reactions and include them as source terms.

2 Governing equations

The flow is treated as incompressible and solved through the incompressible Navier Stokes formulation using the PIMPLE algorithm

$$\nabla \cdot \mathbf{u} = 0 \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\frac{1}{\rho}\nabla P + \nu \nabla^2 \mathbf{u} + \mathbf{g} \tag{2}$$

All active species concentrations are treated by reactive Nernst-Planck transport. Reaction source and potential are iteratively corrected using Picard iterations, iterating the solution over k steps while time is held fixed and only when the solution is internally converged the solution moved forward in time n. Each Picard iteration starts using the potential and equilibrium source from the previous time-step $(\phi^{(1)}, \lambda^{(n)})$

$$\frac{\partial c_i^{(n+1)}}{\partial t} + \nabla \cdot \left(c_i^{(n+1)} \mathbf{u} \right) = \nabla \cdot \left(D_i \nabla c_i^{(n+1)} \right) + \nabla \cdot \left(D_i \frac{z_i F}{RT} c_i^{(n+1)} \nabla \phi^{(k)} \right) + R_i^{(k)} \tag{3}$$

Where c is the concentration of species i, $\mathbf{u}$ is the velocity, D is the diffusion coefficient, z is the charge, F is the Faraday constant, R the universal gas constant, T the temperature, ϕ the potential and R_i the combined contribution of fast and slow reactions

$$R_i^{(k)} = R_{kin.i} + R_{eq.i}^{(k)} \tag{4}$$

Slow reactions are treated kinetically by

$$R_{kin.i} = \sum_j \nu_{ij} \dot{r}_j \tag{5}$$

where ν_{ij} is the stoichiometric constant of species for kinetic reaction j. Fast equilibrium reactions are assumed to instantaneously settle into equilibrium within a time-step and $R_{eq.i}$ determine the required extend of the equilbirum reaction to satisfy local equilibrium

$$R_{eq.i}^{(k)} = \sum_r N_{ir} \lambda_r^{(k)} \tag{6}$$

Where N is the stoichiometric coefficient of the equilibrium reaction r and λ is the equilibrium reaction source/rate. The initial Picard iteration $k = 1$ uses the current potential $\phi^{(n)}$ and no equilibrium correction $\lambda_r = 0$. The solution evaluates the equilibrium residual and checks

$$\mathbf{F}(\mathbf{c}^{(n+1)}) = \mathbf{0}, \quad \text{using} \quad \lambda^{(k)} \tag{7}$$

and if bulk electro-neutrality is satisfied

$$\sum_i z_i c_i = 0, \quad \text{using} \quad \phi^{(k)} \tag{8}$$

if either is violated both are corrected for the next Picard iteration $k + 1$. The required change in equilibrium rate $\Delta\lambda$ is found from

$$\mathbf{J}_\lambda \Delta \boldsymbol{\lambda} = -\mathbf{F} \tag{9}$$

and updated using

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha_\lambda \Delta \boldsymbol{\lambda} \tag{10}$$

The potential is updated from current conservation under constant prescribed current j_{app} Resulting in the current conservation equation

$$\nabla \cdot (\mathbf{J}_D - \kappa \nabla \phi^{(k+1)}) = 0 \tag{11}$$

with diffusive flux

$$\mathbf{j}_D = -F \sum_i z_i D_i \nabla c_i^{(n+1)} \tag{12}$$

and conductivity

$$\kappa = \frac{F^2}{RT} \sum_i z_i^2 D_i c_i^{(n+1)} \tag{13}$$

From which the potential is updated using

$$\mathbf{A}_\phi \Phi = \mathbf{b}_D \tag{14}$$

When converged species are converted to mass fraction

$$Y_i = c_i \frac{M_i}{\rho} \tag{15}$$

and the inert species "I" is explicitly found by

$$Y_I = 1 - \sum_{i \neq I} Y_i \tag{16}$$

pH is the log concentration of H^+ in mol/L

$$pH = -log_{10} \left(\frac{c_{H^+}}{1000} \right) \tag{17}$$

Solving the current conservation with the imposed electro-neutrality constraint makes the potential a Lagrange multiplier problem (see appendix A.3) that can be solved on matrix form as

This gives the updated potential ϕ^{n+1}. The species transport and potential equations are iterated until the concentration and potential fields are mutually consistent and the resulting solution satisfies bulk electro-neutrality. Thereby this becomes a Picard iteration, where the values are updated without advancing time until they are converged, for a schematic overview see section 5.

3 Custom boundary conditions

This section covers the custom boundary condition and helper functions added with this model.

3.1 Electrode boundaries

These boundary condition determines the boundary flux at the electrode patches.

3.1.1 galvanostaticMolarFlux

The concentration gradient at the boundary is formulated as a robin boundary condition consisting of a faradaic diffusion and migration contribution

$$\nabla c_i \cdot \mathbf{n} = \frac{\nu_i j_{app}}{D_i |\nu_e| F} - \frac{z_i F}{RT} c_i \nabla \phi \cdot \mathbf{n} \tag{18}$$

where $\mathbf{n}$ is the unit normal vector, with positive values meaning flux out of the control-volume, ν is the stoichiometric coefficient with positive value meaning production of species i or electrons e and j_{app} is the magnitude of the galvanostaticly applied current, see appendix A.1.1. Special cases for this boundary condition is for non-charged species reaction with the electrode, charged species that does not interact with the electrode and species who are neither charged or interact with the electrodes

$$\nabla c_i \cdot \mathbf{n} = \begin{cases} \frac{\nu_i j_{app}}{D_i |\nu_e| F} & \text{when } z_i = 0 \\ -\frac{z_i F}{RT} c_i \nabla \phi \cdot \mathbf{n} & \text{when } \nu_i = 0 \\ 0 & \text{when } z_i = \nu_i = 0 \end{cases} \tag{19}$$

In the first case only the faradaic and diffusion contributions affect the species, for the second only the migration and for the last a simple zero gradient condition is sufficient.

3.1.2 galvanostaticPotentialFlux

The potential gradient boundary is derived from the current expression (see appendix A.1.1) to a Neumann boundary condition

$$\nabla \phi \cdot \mathbf{n} = \frac{\mathbf{J}_D \cdot \mathbf{n} - j_{n.app}}{\kappa} \tag{20}$$

which determines the potential that balances the diffusive flux and conductivity to the applied current. $j_{n.app}$ is the direction aware current density, see appendix A.1.1.

3.2 Inlet boundaries

These custom boundaries are treats the inlet patch and simplifies some of the commonly used boundary conditions.

3.2.1 pHMolarConcentration

This custom boundary automatically determines the distribution of H^+ and OH^- ions at a given pH. c_{H^+} is directly found from pH using concentration in mol/L

$$c_{H^+} = 10^3 \cdot 10^{-pH} \tag{21}$$

and c_{OH^-} by

$$c_{OH^-} = \frac{K_W}{c_{H^+}} \tag{22}$$

The user points the BC to the relevant equilibrium model from which K_W is read to ensure internal consistency.

3.2.2 inertMolarConcentration

This custom boundary condition is used for the inert species and sets the inlet concentration as the remainder such that the total mass-fraction of the system adds up to 1. The remainder for the inert species "I" is defined

$$Y_I = 1 - \sum_{i \neq I} Y_i \tag{23}$$

Concentration fields are converted to mass-fraction using

$$Y_i = \frac{c_i M_i}{\rho} \tag{24}$$

followed by

$$c_I = \frac{\rho Y_I}{M_I} \tag{25}$$

So directly

$$c_I = \frac{\rho}{M_I} \left(1 - \sum_{i \neq I} \frac{c_i M_i}{\rho} \right) \tag{26}$$

4 Reaction Models

The solver combines all reaction contributions into a single source-term for each species. This combines the contributions of slow kinetic reactions and fast equilibrium reactions (equation 4)

$$R_i = R_{kin.i} + R_{eq.i}$$

4.1 Kinetic Reactions

Kinetic reactions typically develop on the order of or much slower than that of the transport. Therefore they are captured kinetically by their reaction rates (equation 5)

$$R_{kin.i} = \sum_j \nu_{ij} \dot{r}_j$$

Currently the model supports reversible kinetic reactions using known forward and backwards reaction rates and multiplicative Monod used microbial consumption and production.

4.1.1 Reversible kinetic reaction

The reaction rate of reversible kinetic reactions are found as

$$\dot{r} = k_f \prod c_R - k_b \prod c_P \tag{27}$$

where k_f is the forward reaction rate and k_b is the reverse reaction rate. c_R is the concentration of reactants and c_P is the concentration of products. This model can allow for negative concentrations when used for fast equilibrium models.

4.1.2 Multiplicative Monod reaction

Microbial reactions are described through multiplicative Monod

$$\dot{r} = X \dot{q}_{max} \left(\frac{c_d}{c_d + K_{S.d}} \right) \left(\frac{c_a}{c_a + K_{S.a}} \right) \tag{28}$$

where X is the biomass density, $\dot{q}_{max}$ is the maximum specific consumption rate, K_S is the half saturation constant and the subscripts d and a describe the electron donor and acceptor substrates.

4.2 Equilibrium Reactions

Fast equilibrium reactions that are much faster than that of transport and slow kinetic reactions can be assumed to instantaneously settle into equilibrium within each solver time-step. This allows for treatment of fast reactions without resolving the kinetic rates which becomes very stiff for fast reactions. Instead they are incorporated as an equilibrium source term for each species telling how far the equilibrium reaction must proceed to satisfy the equilibrium condition of the system (equation 6)

$$R_{eq.i} = \sum_r N_{ir} \lambda_r$$

This is done for a general setup supporting a system of fast equilibrium reactions. For a given set of equilibrium reactions, all participating species are collected in the concentration vector of N species

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{bmatrix} \tag{29}$$

The combined stoichiometric coefficients for all species N of all reactions r is defined

$$\mathbf{N} = \begin{bmatrix} \nu_{11} & \nu_{12} & \cdots & \nu_{1r} \\ \nu_{21} & \nu_{22} & \cdots & \nu_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ \nu_{N1} & \nu_{N2} & \cdots & \nu_{Nr} \end{bmatrix} \tag{30}$$

From the equilibrium reaction

$$K = \frac{\prod c_P^{\nu}}{\prod c_R^{\nu}} \tag{31}$$

The equilibrium residual equation system is formed

$$\mathbf{F}(\mathbf{c}) = \begin{bmatrix} \prod c_{p.1}^{\nu} - K_1 \prod c_{R.1}^{\nu} \\ \prod c_{p.2}^{\nu} - K_2 \prod c_{R.2}^{\nu} \\ \vdots \\ \prod c_{p.r}^{\nu} - K_r \prod c_{R.r}^{\nu} \end{bmatrix} = \mathbf{0} \tag{32}$$

From this the concentration Jacobian is defined

$$\mathbf{J}_c = \frac{\partial \mathbf{F}}{\partial \mathbf{c}} \tag{33}$$

Using the local transport response $\mathbf{S}$ (see appendix A.2.1) this becomes the equilibrium Jacobian

$$\mathbf{J}_\lambda = \mathbf{J}_c \mathbf{S} \mathbf{N} \tag{34}$$

Then the local equilibrium problem simplifies to

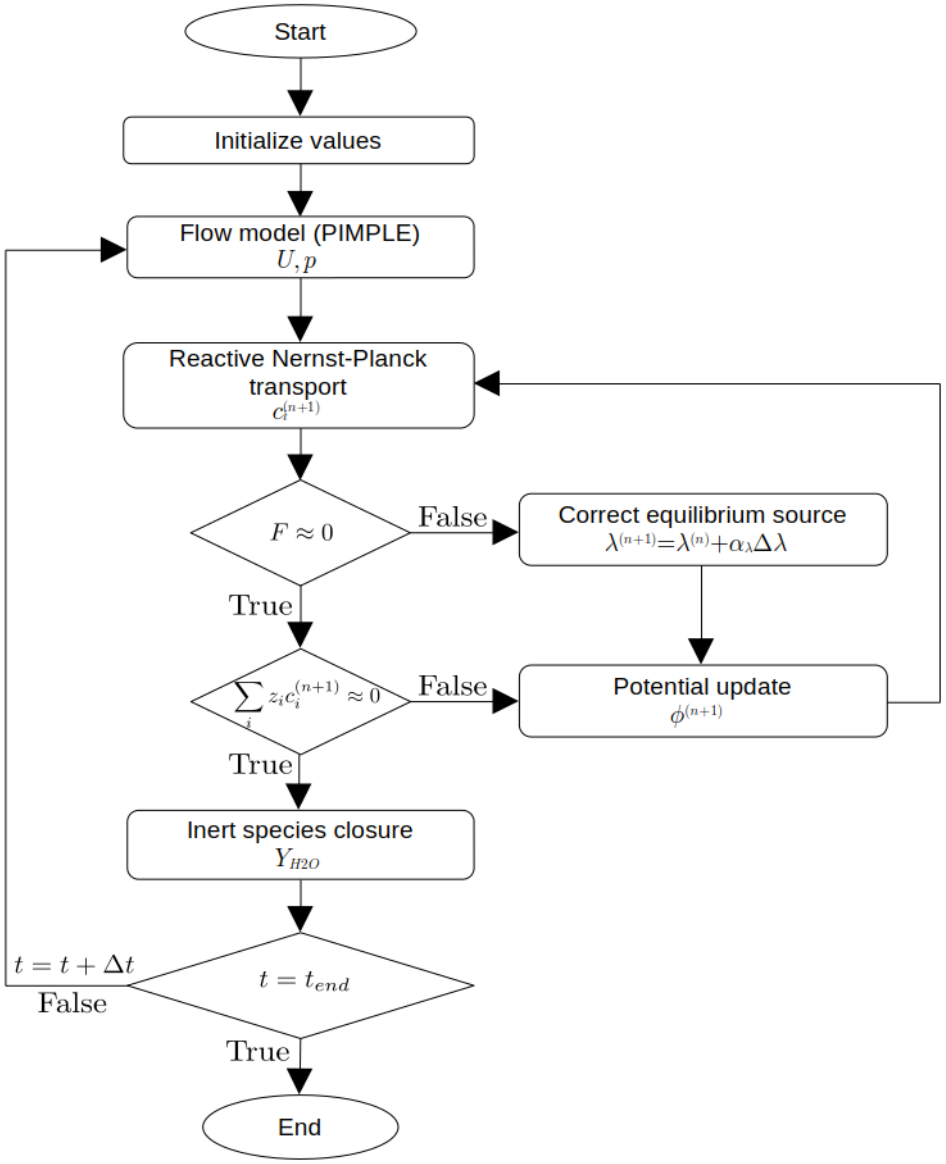
$$\mathbf{J}_\lambda \Delta \boldsymbol{\lambda} = -\mathbf{F} \tag{35}$$

See appendix A.2.1. From this the Newton correction $\Delta\boldsymbol{\lambda}$ becomes the Lagrange multiplier to the linear matrix system that determines how far the equilibrium source must be corrected to satisfy equilibrium

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha_\lambda \Delta \boldsymbol{\lambda} \tag{36}$$

where α_λ is a relaxation factor ($0 \leq \alpha_\lambda \leq 1$).

5 Algorithm flowchart



6 Solver structure

Source code structure

```
electroChem/
|-- Documentation/
|-- Make/
|   |-- files
|   |-- options
|-- reactionModels/
|   |-- reactionManager.C
|   |-- reactionManager.H
|   |-- equilibriumReactions/
|   |   |-- fastEquilibrium.C
|   |   |-- fastEquilibrium.H
|   |-- kineticReactions/
|       |-- multiplicativeMonod/
|       |   |-- multiplicativeMonod.C
|       |   |-- multiplicativeMonod.H
|       |-- reversibleKineticReaction/
|           |-- reversibleKineticReaction.C
|           |-- reversibleKineticReaction.H
|-- cEqn.H
|-- createFields.H
|-- diagnostics.H
|-- electroChem.C
|-- pEqn.H
|-- phiEqn.H
|-- UEqn.H
|-- YEqn.H
```

custom functions

```
src/
|-- customLibs/
|   |-- electrochemistry/
|       |-- derivedFvPatchFields/
|           |-- MolarFlux/
|           |   |-- galvanostaticMolarFlux.C
|           |   |-- galvanostaticMolarFlux.H
|           |-- PotentialFlux/
|           |   |-- galvaniostaticPotentialFlux.C
|           |   |-- galvaniostaticPotentialFlux.H
|           |-- MolarConcentration/
|               |-- pHMolarConcentration.C
|               |-- pHMolarConcentration.H
|               |-- inertMolarConcentration.C
|               |-- inertMolarConcentration.H
|-- Make/
    |-- files
    |-- options
```

case file

```
case/
|-- 0/
|   |-- U
|   |-- P
|   |-- PhiE
|   |-- c_*
|-- constant/
|   |-- electroChemistry
|   |-- reactions
|   |-- transportProperties
|   |-- turbulenceProperties
|-- system/
|   |-- controlDict
|   |-- fvSchemes
|   |-- fvSolution
```

A Appendix

A.1 Boundary conditions

This section relies on the following sign conventions

$$\begin{aligned}
\mathbf{n}: &\quad \text{control-volume outward-pointing unit normal vector} \\
j_{app}: &\quad \text{magnitude of applied current density,} \quad j_{app} \geq 0 \\
\mathbf{N}_i \cdot \mathbf{n} > 0: &\quad \text{species flux out of control-volume} \\
\mathbf{j} \cdot \mathbf{n} > 0: &\quad \text{ionic current out of control-volume} \\
\\
\nu_i > 0: &\quad \text{production of aqueous species } i \\
\nu_i < 0: &\quad \text{consumption of aqueous species } i \\
\nu_e > 0: &\quad \text{production of electrons} \\
\nu_e < 0: &\quad \text{consumption of electrons}
\end{aligned}$$

A.1.1 Electrode boundaries

For a species "i" with flux

$$\mathbf{N}_i = \mathbf{u}c_i - D_i \nabla c_i - \frac{z_i F}{RT} D_i c_i \nabla \phi \tag{37}$$

At an impermeable solid electrode $\mathbf{u} \cdot \mathbf{n} = 0$, the surface normal flux is

$$\mathbf{N}_i \cdot \mathbf{n} = -D_i \nabla c_i \cdot \mathbf{n} - \frac{z_i F}{RT} D_i c_i \nabla \phi \cdot \mathbf{n} \tag{38}$$

OpenFOAM sign conventions follows that $\mathbf{n} > 0$ describes flux out of the control volume and if a positive stoichiometric coefficient $\nu_i > 0$ means production of species i then the faradaic production at an electrode must follow

$$N_{i.e} = -\frac{\nu_i j_{app}}{|\nu_e| F} \tag{39}$$

where j_{app} is the magnitude of the galvanostaticly applied current, and ν_e is the stoichiometric coefficient for electrons participating in the reaction. Hence, if $\nu_i > 0$ then $N_{i.e} < 0$ and species is produced into the control-volume and oppositely when $\nu_i < 0$ species is consumed. This faradic flux, equations 39, must be consistent with the Nernst-Planck flux, equation 38

$$\mathbf{N}_i \cdot \mathbf{n} = N_{i.e} \tag{40}$$

Concentration gradient

To get the concentration gradient at the boundary equations 38 and 39 are inserted into equation 40

$$-D_i \nabla c_i \cdot \mathbf{n} - \frac{z_i F}{RT} D_i c_i \nabla \phi \cdot \mathbf{n} = -\frac{\nu_i j_{app}}{|\nu_e| F} \tag{41}$$

In this $\nabla c_i \cdot \mathbf{n}$ is isolated to form a robin boundary condition

$$\nabla c_i \cdot \mathbf{n} = \frac{\nu_i j_{app}}{D_i |\nu_e| F} - \frac{z_i F}{RT} c_i \nabla \phi \cdot \mathbf{n} \tag{42}$$

Discretization

The boundary is solved using a Robin equation formulation. Introduce

$$A = \frac{\nu_i j_{app}}{|\nu_e| F D_i}, \quad \beta = \frac{z_i F}{RT} (\nabla \phi \cdot \mathbf{n}) \tag{43}$$

At the electrode boundary

$$\nabla c_i \cdot \mathbf{n} = A - \beta c_{b.i} \tag{44}$$

where $c_{b.i}$ is the boundary concentration of species i. Then the boundary adjacent gradient is defined as

$$\nabla c_i = \delta(c_{b.i} - c_{p.i}) \tag{45}$$

where $\delta = 1/\Delta_n$, Δ_n is the distance from boundary to cell-center and $c_{p.i}$ is the boundary adjacent cell-center concentration. Insert into equation 44

$$\delta(c_{b.i} - c_{p.i}) = A - \beta c_{b.i} \tag{46}$$

isolate boundary concentration

$$c_{b.i} = \frac{\delta c_{p.i} + A}{\delta + \beta} \tag{47}$$

then the boundary flux becomes

$$\nabla c_i \cdot \mathbf{n} = A - \beta c_{b.i} \tag{48}$$

Potential gradient

The potential gradient is derived from the current

$$\mathbf{j} = F \sum_i z_i \mathbf{N}_i \tag{49}$$

on electrode normal component form

$$\mathbf{j} \cdot \mathbf{n} = F \sum_i z_i (\mathbf{N}_i \cdot \mathbf{n}) \tag{50}$$

From equation 40 it is given that $\mathbf{N}_i \cdot \mathbf{n} = N_{i.e}$ which is inserted

$$\mathbf{j} \cdot \mathbf{n} = F \sum_i z_i N_{i.e} \tag{51}$$

which expands to

$$\mathbf{j} \cdot \mathbf{n} = -\frac{j_{app}}{|\nu_e|} \sum_i z_i \nu_i \tag{52}$$

From the charge balance of the electrode half-reaction, given the electrode charge is fixed as $z_e = -1$ then

$$\sum_i z_i \nu_i + z_e \nu_e = 0 \Rightarrow \sum_i z_i \nu_i = \nu_e \tag{53}$$

This is substituted back into equation 52

$$\mathbf{j} \cdot \mathbf{n} = -\frac{\nu_e}{|\nu_e|} j_{app} \tag{54}$$

This shows the direction of the electrode normal current follows from the electron stoichiometry of the half-reaction. or an electron-consuming reaction, ($\nu_e < 0$), the normal current is positive and conventional ionic current leaves the electrolyte through the electrode boundary. For an electron-producing reaction, ($\nu_e > 0$), the normal current is negative and conventional ionic current enters the electrolyte through the electrode boundary. For simplicity an electrode aware current density is defined as

$$j_{n.app} \equiv \mathbf{j} \cdot \mathbf{n} = -\frac{\nu_e}{|\nu_e|} j_{app} \tag{55}$$

Expanding equation 49 with 38 gives the current equation

$$\mathbf{j} = \mathbf{J}_D - \kappa \nabla \phi \tag{56}$$

with diffusive flux

$$\mathbf{J}_D = -F \sum_i z_i D_i \nabla c_i \tag{57}$$

and conductivity

$$\kappa = \frac{F^2}{RT} \sum_i z_i^2 D_i c_i \tag{58}$$

This can be rewritten as

$$(\mathbf{J}_D - \kappa \nabla \phi) \cdot \mathbf{n} = j_{n.app} \tag{59}$$

where the potential gradient can be isolated to a Neumann boundary condition

$$\nabla \phi \cdot \mathbf{n} = \frac{\mathbf{J}_D \cdot \mathbf{n} - j_{n.app}}{\kappa} \tag{60}$$

A.2 Fast equilibrium reactions

A.2.1 Local approximation

Starting from transport

$$\frac{\partial c_i^{(n+1)}}{\partial t} + \nabla \cdot \left(c_i^{(n+1)} \mathbf{u} \right) = \nabla \cdot \left(D_i \nabla c_i^{(n+1)} \right) + \nabla \cdot \left(D_i \frac{z_i F}{RT} c_i^{(n+1)} \nabla \phi^{(n)} \right) + R_{kin.i} + R_{eq.i} \tag{61}$$

$$\mathcal{T}_i(c_i, \phi) = R_{kin.i} + \sum_r N_{ir} \lambda_r \tag{62}$$

with constraint

$$\mathbf{F}(\mathbf{c}) = 0 \tag{63}$$

where $\mathcal{T}_i$ is the transient and Nernst-Planck species transport and λ_r is the unknown reaction source term required to satisfy equilibrium through transport. On discretized form over the domain this is represented as

$$\mathbf{A}_i \mathbf{c}_i = \mathbf{b}_i + \mathbf{V} \sum_r N_{ir} \boldsymbol{\lambda}_r \tag{64}$$

where $\mathbf{A}_i$ is the assembled discrete transport operator, $\mathbf{b}_i$ is the known discrete contributions and the third term is the equilibrium source correction. The detailed coefficients in $\mathbf{A}_i$ and $\mathbf{b}_i$ depend on the chosen temporal, convection, gradient and Laplacian schemes and are not expanded here. Using this expression the equilibrium source is perturbed by $\Delta \boldsymbol{\lambda}_r$

$$\boldsymbol{\lambda}_r \to \boldsymbol{\lambda}_r + \Delta \boldsymbol{\lambda}_r \tag{65}$$

The concentration will respond accordingly by

$$\mathbf{c}_i \to \mathbf{c}_i + \Delta \mathbf{c}_i \tag{66}$$

If $\mathbf{A}_i$ and $\mathbf{b}_i$ are held fixed during the source-correction step, then

$$\mathbf{A}_i (\mathbf{c}_i + \Delta \mathbf{c}_i) = \mathbf{b}_i + \mathbf{V} \sum_r N_{ir} (\boldsymbol{\lambda}_r + \Delta \boldsymbol{\lambda}_r) \tag{67}$$

Subtracting the current system (equation 64) gives

$$\mathbf{A}_i \Delta \mathbf{c}_i = \mathbf{V} \sum_r N_{ir} \Delta \boldsymbol{\lambda}_r \tag{68}$$

Evaluating this for every equilibrium-source Newton correction would require non-local transport solves. Therefore this is simplified by only considering local cell-center values. Effectively for a cell P this means only retaining the diagonal responses of the transport operator

$$a_{P.i} \Delta c_{P.i} \approx V_P \sum_r N_{ir} \Delta \lambda_{P.r} \tag{69}$$

isolating $\Delta c_{P.i}$ gives

$$\Delta c_{P.i} \approx \frac{V_P}{a_{P.i}} \sum_r N_{ir} \Delta \lambda_{P.r} \tag{70}$$

From this we combine the constants by defining

$$s_{P.i} = \frac{V_P}{a_{P.i}} \tag{71}$$

And over all cells N

$$\mathbf{S}_P = \mathrm{diag}(s_{p.1}, ..., s_{P.N}) \tag{72}$$

then the local concentration response becomes

$$\Delta \mathbf{c}_P \approx \mathbf{S}_P \mathbf{N} \Delta \boldsymbol{\lambda}_P \tag{73}$$

General solution form

For a given set of equilibrium reactions, all participating species are collected in the concentration vector of N species

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{bmatrix} \tag{74}$$

The combined stoichiometric coefficients for all species N of all reactions r is defined

$$\mathbf{N} = \begin{bmatrix} \nu_{11} & \nu_{12} & \cdots & \nu_{1r} \\ \nu_{21} & \nu_{22} & \cdots & \nu_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ \nu_{N1} & \nu_{N2} & \cdots & \nu_{Nr} \end{bmatrix} \tag{75}$$

From the equilibrium reaction

$$K = \frac{\prod c_P^{\nu}}{\prod c_R^{\nu}} \tag{76}$$

The equilibrium residual equation system is formed

$$\mathbf{F}(\mathbf{c}) = \begin{bmatrix} \prod c_{p.1}^{\nu} - K_1 \prod c_{R.1}^{\nu} \\ \prod c_{p.2}^{\nu} - K_2 \prod c_{R.2}^{\nu} \\ \vdots \\ \prod c_{p.r}^{\nu} - K_r \prod c_{R.r}^{\nu} \end{bmatrix} = \mathbf{0} \tag{77}$$

then the concentration Jacobian is defined

$$\mathbf{J}_c = \frac{\partial \mathbf{F}}{\partial \mathbf{c}} = \begin{bmatrix} \frac{\partial F_1}{\partial c_1} & \frac{\partial F_1}{\partial c_2} & \cdots & \frac{\partial F_1}{\partial c_N} \\ \frac{\partial F_2}{\partial c_1} & \frac{\partial F_2}{\partial c_2} & \cdots & \frac{\partial F_2}{\partial c_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_r}{\partial c_1} & \frac{\partial F_r}{\partial c_2} & \cdots & \frac{\partial F_r}{\partial c_N} \end{bmatrix} \tag{78}$$

The equilibrium residuals are expanded using first order Taylor-expansion ($f(x + \Delta x) \approx f(x) + \frac{df}{dx}\Delta x$)

$$\mathbf{F}(\mathbf{c} + \Delta\mathbf{c}) \approx \mathbf{F}(\mathbf{c}) + \mathbf{J}_c \Delta\mathbf{c} \tag{79}$$

This expression is constrained such equilibrium must be satisfied

$$\mathbf{F} + \mathbf{J}_c \Delta\mathbf{c} \approx 0 \tag{80}$$

In this the local transport response (equation 73) is substituted

$$\mathbf{J}_c \mathbf{S}\mathbf{N} \Delta\boldsymbol{\lambda} = -\mathbf{F} \tag{81}$$

Now define the equilibrium Jacobian

$$\mathbf{J}_\lambda = \mathbf{J}_c \mathbf{S}\mathbf{N} \tag{82}$$

Then the local equilibrium problem simplifies to

$$\mathbf{J}_\lambda \Delta\boldsymbol{\lambda} = -\mathbf{F} \tag{83}$$

from which the Newton correction $\Delta\boldsymbol{\lambda}$ becomes the Lagrange multiplier to the linear matrix system that determines how far the equilibrium source must be corrected to satisfy equilibrium

$$\boldsymbol{\lambda}^{(n+1)} = \boldsymbol{\lambda}^{(n)} + \alpha_\lambda \Delta\boldsymbol{\lambda} \tag{84}$$

where α_λ is a relaxation factor ($0 \leq \alpha_\lambda \leq 1$).

A.3 Potential

A.3.1 Lagrange equation derivation

Potential is found through current conservation with current equation

$$\mathbf{j} = F \sum_i z_i \mathbf{N}_i \tag{85}$$

Where "$\mathbf{N}_i$" is the species flux

$$\mathbf{N}_i = \mathbf{u}c_i - D_i \nabla c_i - \frac{z_i F}{RT} D_i c_i \nabla \phi \tag{86}$$

and charge density

$$\rho_e = F \sum_i z_i c_i \tag{87}$$

Including reaction the full current conservation equation becomes

$$\frac{\partial \rho_e}{\partial t} + \nabla \cdot \mathbf{j} = F \sum_i z_i R_i \tag{88}$$

This expands to

$$\frac{\partial \rho_e}{\partial t} + \nabla \cdot \left(F \mathbf{u} \sum_i z_i c_i \right) - \nabla \cdot \left(F \sum_i z_i D_i \nabla c_i \right) - \nabla \cdot \left(\frac{F^2}{RT} \sum_i z_i^2 D_i c_i \nabla \phi \right) = F \sum_i z_i R_i \tag{89}$$

From this potential can be solved such that current is conserved, however a solution that additionally enforce electro-neutrality is desired. For this an additional constraint of no charge concentration is imposed

$$\rho_e = 0 \Rightarrow \sum_i z_i c_i = 0 \tag{90}$$

Additionally as reactions are charge conserving the following can be imposed too

$$\sum z_i R_i = 0 \tag{91}$$

With this the charge density, advection and reaction terms become zero and the current conservation simplifies to a divergence free current equation

$$\nabla \cdot \mathbf{j} = 0 \Rightarrow \nabla \cdot \left(-F \sum z_i D_i \nabla c_i - \frac{F^2}{RT} \sum z_i^2 D_i c_i \nabla \phi \right) = 0 \tag{92}$$

From this diffusive flux is defined as

$$\mathbf{J}_D = -F \sum z_i D_i \nabla c_i \tag{93}$$

and conductivity

$$\kappa = \frac{F^2}{RT} \sum z_i^2 D_i c_i \tag{94}$$

Substituting these into the current conservation equation gives

$$\nabla \cdot (\mathbf{J}_D - \kappa \nabla \phi) = 0 \tag{95}$$

or rearranged to elliptic form

$$\nabla \cdot (\kappa \nabla \phi) = \nabla \cdot \mathbf{J}_D \tag{96}$$

The electric potential acts as the Lagrange multiplier associated with the bulk electro-neutrality constraint. Requiring the species evolution to preserve electro-neutrality yields the divergence-free current condition, from which the elliptic potential equation is obtained.

A.3.2 Discretization

To solve this using FVM the equation must be discretized. for a volume cell V_P with cell-center P the integral form of the equation is

$$\int_{V_P} \nabla \cdot (\kappa \nabla \phi)\, dV = \int_{V_P} \nabla \cdot \mathbf{J}_D\, dV \tag{97}$$

Applying Gauss' divergence theorem converts to a surface integral

$$\oint_{\partial V_P} (\kappa \nabla \phi) \cdot \mathbf{n} dA = \oint_{\partial V_P} \mathbf{J}_D \cdot \mathbf{n} dA \tag{98}$$

where ∂V_P is the boundary surface of the control volume. This converts the differential conservation equation into a balance of fluxes through the control-volume boundary. Approximating the surface integral as a sum over the cell faces "f" gives

$$\sum_{f \in P} \kappa_f (\nabla \phi)_f \cdot \mathbf{S}_f = \sum_{f \in P} \mathbf{J}_{D.f} \cdot \mathbf{S}_f \tag{99}$$

where $\mathbf{S}_f = A_f \mathbf{n}_f$ is the face area vector. To illustrate how this is decomposed into a linear matrix system, an orthogonal grid is considered where face "f" connects cell P to its neighbour N. The potential gradient at the face can be approximated using

$$(\nabla \phi) \cdot \mathbf{S}_f \approx \frac{\phi_N - \phi_P}{d_{PN}} A_f \tag{100}$$

substituting this into the finite volume representation gives

$$\sum_N \kappa_f \frac{\phi_N - \phi_P}{d_{PN}} A_f = \sum_{f \in P} \mathbf{J}_{D.f} \cdot \mathbf{S}_f \tag{101}$$

The constants are combined into

$$a_{PN} = \frac{\kappa_f A_f}{d_{PN}} \tag{102}$$

So the expression becomes

$$\sum_N a_{PN} (\phi_N - \phi_P) = \sum_{f \in P} \mathbf{J}_{D.f} \cdot \mathbf{S}_f \tag{103}$$

By expanding and multiplying by -1 the following is obtained

$$\phi_P \sum_N a_{PN} - \sum_N a_{PN} \phi_N = -\sum_{f \in P} \mathbf{J}_{D.f} \cdot \mathbf{S}_f \tag{104}$$

The net diffusion flux of the cell on the right hand side is defined as

$$b_{D.P} = -\sum_{f \in P} \mathbf{J}_{D.f} \cdot \mathbf{S}_f \tag{105}$$

and the expression becomes

$$\phi_P \sum_N a_{PN} - \sum_N a_{PN} \phi_N = b_{D.P} \tag{106}$$

Inserting the values for each cell 1,...,n

$$\phi_1 \sum_N a_{PN} - \sum_N a_{PN} \phi_N = b_{D.1}$$

$$\phi_2 \sum_N a_{PN} - \sum_N a_{PN} \phi_N = b_{D.2}$$

$$\vdots$$

$$\phi_n \sum_N a_{PN} - \sum_N a_{PN} \phi_N = b_{D.n}$$

on matrix form this becomes

$$\mathbf{A}_\phi \Phi = \mathbf{b}_D \tag{107}$$

with

$$\mathbf{A}_\phi = \begin{bmatrix} \sum_N a_{PN} & -a_{12} & \cdots \\ -a_{21} & \sum_N a_{PN} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}, \quad \Phi = \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_n \end{bmatrix}, \quad \mathbf{b}_D = \begin{bmatrix} b_{D.1} \\ b_{D.2} \\ \vdots \\ b_{D.n} \end{bmatrix} \tag{108}$$

From this the Φ is solved as the Lagrange multiplier that determines the potential such that current is conserved in a way that satisfies electro-neutrality. For non-orthogonal grids the same method is applied but the gradiant approximation will differ depending on the cell relations.

A.4 MES setup

A.4.1 Reaction simplifications

Full reaction scheme

$$\begin{aligned} r_W: &\quad H_2O \rightleftharpoons H^+ + OH^- \\ r_1: &\quad \mathrm{CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+} \\ r_2: &\quad HCO_3^- \rightleftharpoons CO_3^{2-} + H^+ \\ r_3: &\quad CO_2 + OH^- \rightleftharpoons HCO_3^- \\ r_4: &\quad CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^- \end{aligned}$$

The CO_2-bicarbonate equilibrium r_1 consists of a lumped hydration step $r_{1.h}$ and acid dissociation step $r_{1.a}$

$$r_1: \mathrm{CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+} \begin{cases} r_{1.h}: & \mathrm{CO_2 + H_2O \rightleftharpoons H_2CO_3} \\ r_{1.a}: & \mathrm{H_2CO_3 \rightleftharpoons HCO_3^- + H^+} \end{cases}$$

The CO_2 hydration step as well as the hydroxide path r_3 are both slow reactions and cannot comfortably be treated as instantaneous equilibrium projections and must be treated kinetically

$$\dot{r}_{1.h} = k_{f.1} c_{CO_2} - k_{b.1} c_{H_2CO_3} \tag{109}$$

$$\dot{r}_3 = k_{f.3} c_{CO_2} c_{OH^-} - k_{b.3} c_{HCO_3^-} \tag{110}$$

reactions $r_{1.h}$ and $r_{1.a}$ can be thermodynamically lumped into a single kinetic equation.

$$K_{1.2} = \frac{c_{HCO_3^-} c_{H^+}}{c_{H_2CO_3}} \Rightarrow c_{H_2CO_3} = \frac{c_{HCO_3^-} c_{H^+}}{K_{1.2}} \tag{111}$$

$$\dot{r}_1 = k_{f.1} c_{CO_2} - \frac{k_{b.1}}{K_{1.2}} c_{HCO_3^-} c_{H^+} \tag{112}$$

Effectively eliminating H_2CO_3 as a variable and making CO_2 treated entirely kinetically. Then the renaming equilibrium reactions are

$$K_W = c_{H^+} c_{OH^-} \tag{113}$$

$$K_2 = \frac{c_{CO_3^{2-}} c_{H^+}}{c_{HCO_3^-}} \tag{114}$$

$$K_4 = \frac{c_{HCO_3^-} c_{OH^-}}{c_{CO_3^{2-}}} \tag{115}$$

Where r_4 can be shown to be redundant. The equilibrium equations for water and bicarbonate-carbonate are rewritten

$$K_2 = \frac{c_{CO_3^{2-}} c_{H^+}}{c_{HCO_3^-}} \Rightarrow \frac{c_{HCO_3^-}}{c_{CO_3^{2-}}} = \frac{c_{H^+}}{K_2} \tag{116}$$

$$K_W = c_{H^+} c_{OH^-} \Rightarrow c_{OH^-} = \frac{K_W}{c_{H^+}} \tag{117}$$

The rewritten forms are multiplied and H^+ cancel out on the right-hand side

$$\frac{c_{HCO_3^-} c_{OH^-}}{c_{CO_3^{2-}}} = \frac{c_{H^+} K_W}{K_2 c_{H^+}} = \frac{K_W}{K_2} \tag{118}$$

Showing the left-hand side is exactly the equilibrium constant for r_4

$$K_4 = \frac{K_W}{K_2} \tag{119}$$

making it a linear combination of the other equilibrium reactions and thereby redundant as they are not resolved kinetically but projected.

A.4.2 Equilibrium Newton system

From the fast equilibrium reactions

$$H_2O \rightleftharpoons H^+ + OH^- \tag{120}$$

$$HCO_3^- \rightleftharpoons CO_3^{2-} + H^+ \tag{121}$$

the concentration vector is defined

$$\mathbf{c} = \begin{bmatrix} c_{H^+} \\ c_{OH^-} \\ c_{HCO_3^-} \\ c_{CO_3^{2-}} \end{bmatrix} \tag{122}$$

and using reaction ordering (r_W, r_2), then the stoichiometric matrix becomes

$$\mathbf{N} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{bmatrix} \tag{123}$$

and constraint equations

$$\mathbf{F}(\mathbf{c}) = \begin{bmatrix} c_{H^+}c_{OH^-} - K_W \\ c_{CO_3^{2-}}c_{H^+} - K_2 c_{HCO_3^-} \end{bmatrix} \tag{124}$$

then the concentration Jacobian is defined

$$\mathbf{J}_c = \frac{\partial \mathbf{F}}{\partial \mathbf{c}} = \begin{bmatrix} c_{OH^-} & c_{H^+} & 0 & 0 \\ c_{CO_3^{2-}} & 0 & -K_2 & c_{H^+} \end{bmatrix} \tag{125}$$